

Introduction to superconductivity.

superconductivity is a phenomenon occurring in certain material at very low temp. characterized by exactly zero electrical resistance & the exclusion of the interior magnetic field.

It was 1st discovered by H. Kameeligh Onnes in 1911.

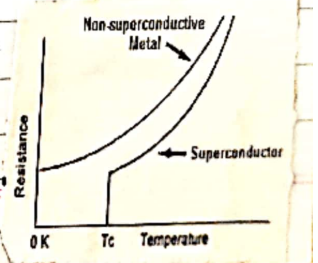
- The electrical resistivity of a metallic conductor decreases gradually as the temp. lowered.
- However, in ordinary conductors such as copper & silver, impurities & other defects impose a lower limit. Even near absolute zero a real sample of copper shows non zero resistance.
- The resistance of superconductor, ~~on~~ drops abruptly to zero when the material is cooled below its "critical temp."
- H. Kameeligh Onnes, successfully liquified helium.
- As helium boils at 4.2 K, it therefore possible to study the properties of material at low temp.
- In 1911, he observed that electrical resistivity of pure mercury dropped suddenly to zero at about the boiling point of helium.
- He ~~conclude~~ concluded that mercury had passed in the new state, which he called the superconducting state.

def: T_c := This is critical temp. at which the

resistivity of a superconductor goes to zero.

Above this temp. the material is non-superconducting while below it, the material becomes superconducting.

superconductor := A conductor having zero or almost zero resistance is called a superconductor.



Properties of superconductor :-

1) Zero electrical resistance :-

- A superconductor is characterized by zero resistivity. Experimentally it is not possible to test resistance is zero.
- Zero resistance can be test by one method in which material is connected in the circuit & Potential difference is measured across the end.

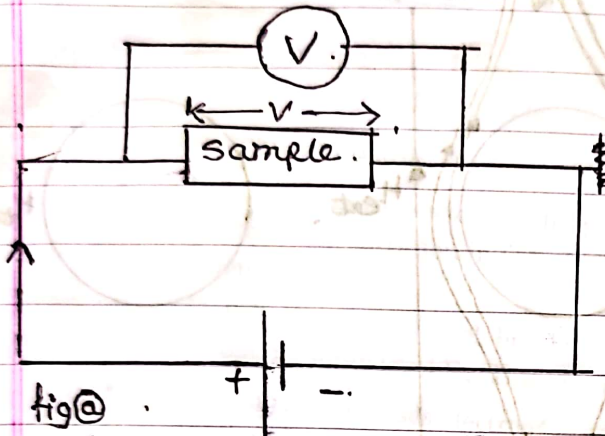
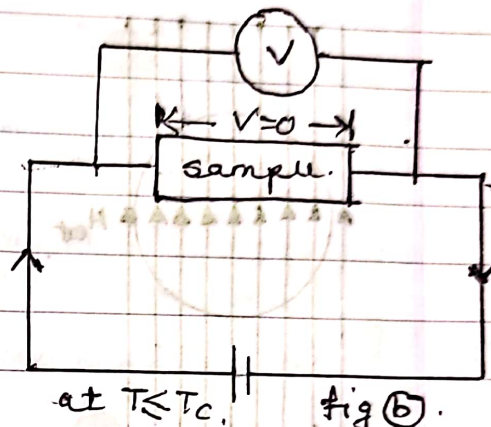


fig (a) at $T > T_c$, $\frac{V}{I} = R$ [Normal state]



at $T \leq T_c$, $\frac{V}{I} = 0$ [superconducting state] fig (b)

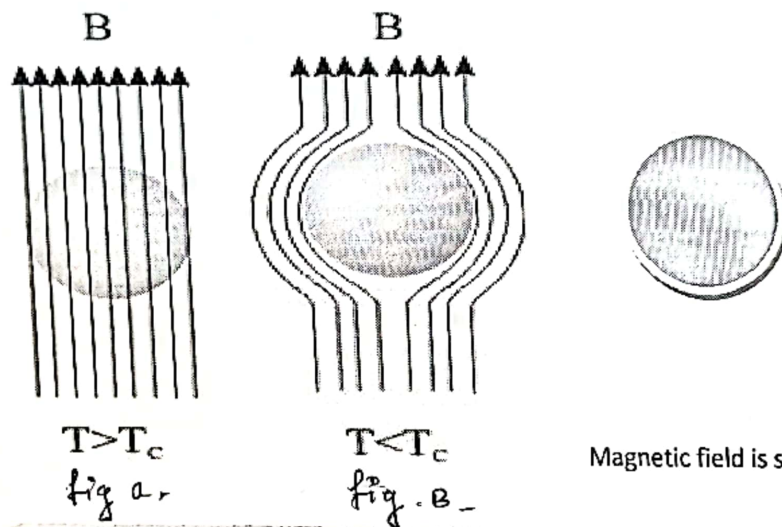
- When the material is in normal conducting state a voltage drop is observed across the ends of the material as shown in fig (a).
- When the material is in the superconducting state i.e. when cooled below its critical temp. T_c , the potential difference across the ends of the material shows sudden drop to zero as shown in fig (b).

2) Maisner's effect :-

- In 1933 Meissner discovered that a superconductor completely expels any magnetic field lines that were initially penetrating it in its normal state.
- This phenomenon states that ^{when} a superconductor is cooled below critical temp. T_c & placed in magnetic field H then the flux is excluded from the bulk of the specimen.
- Suppose the magnetic field is applied $\perp$ to the

sample in the normal state as shown in fig a. then magnetic flux can pass through the specimen.

- Let the material be cooled below the critical temp. T_c in the presence of magnetic field. then the magnetic flux was totally excluded from the sample as it becomes ~~conducts~~ superconductor as it becomes shown in fig b.



- Hence, the expulsion of magnetic flux during the transition from normal to superconducting state is known as Meissner effect.
- When the magnetic field is switched off, magnetic field is not trapped by superconductor as shown in fig c.
- The magnetic induction inside the specimen in normal state is given by -

$$B = \mu_0 (H + M)$$

Normal state.

where H = external applied field.
 M = magnetisation produced in specimen.
- According to Meissner's effect, at $T < T_c$, $B = 0$

$$\therefore \mu_0 (H + M) = 0$$

Superconducting state.

$M = -H$

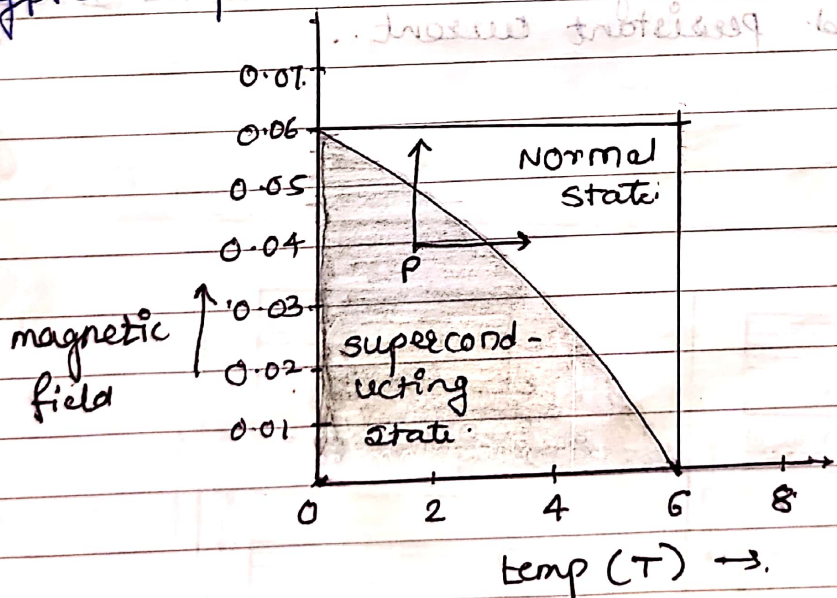
BES Theory

The susceptibility of material is.

$$\chi = \frac{M}{H} = -1 \quad \text{of perfect diamagnetic}$$

Critical magnetic field :-

- K. Onnes discovered in 1913 that, when a superconductor is placed in an increasing magnetic field, it loses superconductivity at a certain value H_c of the field. The magnetic field strength at which superconductivity gets destroyed is called 'critical magnetic field (H_c)'.
- This value is a characteristic of the metal & depends on its orientation in the mag. field & temp.
- The value of H_c varies with temp.
- Fig. shows the variation of H_c with temp for a typical superconductor.



- From fig. consider pt. P, where the temp. & magnetic field are within shaded region, the metal is in superconducting state.
- On increasing either the temp or the field, then it can be driven into normal state.

The variation of critical field with temp is given by relation.

$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right) \right]^2.$$

where, $H_c(0)$ is critical magnetic field at 0°K .

Persistent current :-

consider a superconducting ring placed in a magnetic field. When it cooled below the critical temp., it becomes superconducting.

The external field induces a current in the ring. When ^{Field} switched off, the current will continue to keep flowing on its own accord, around the loop, as long as the loop is held below critical temp. Such a steady current flowing with undiminished strength is called persistent current.

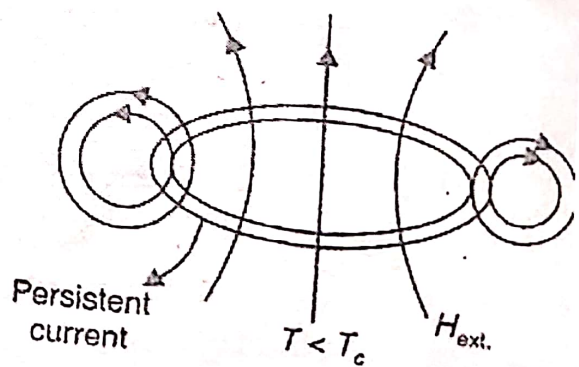
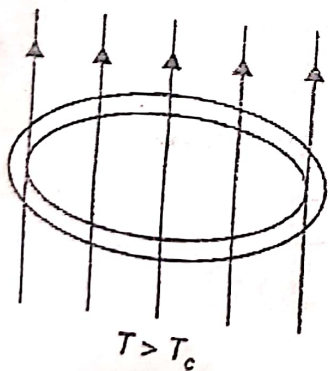


FIGURE 11.31 Persistent Current

Isotope effect :-
 Maxwell & Reynolds found independently that the critical temp. of superconductor varies with isotopic mass.

$$T_c \propto \frac{1}{\sqrt{M}}$$

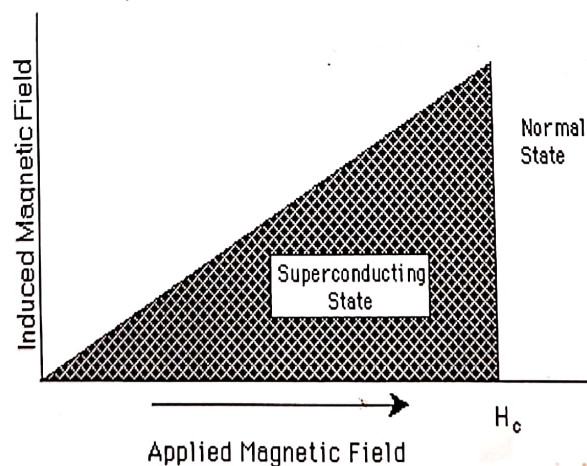
$$T_c \propto M^{-1/2}$$

Types of superconductor.

1) Type - I Superconductor :=

- In type-I superconductor, the transition from a superconducting state to normal state in presence of mag. field, occurs sharply at the critical value H_c . At this point, the field penetrates completely.
- Below H_c , type-I superconductors are perfectly diamagnetic.
- They completely expel the magnetic field from the interior of the specimen.
- Upto the critical field strength, magnetization of the material grows in proportion to the external field.
- At the transition temp., it suddenly drops to zero to the normal conducting state. The magnetic field penetrates only the surface layer & current flows only in this layer.
- e.g. Aluminium & lead.
- Type I superconductor is also called as soft superconductor.

Type I Superconductor



2) Type II Superconductor :=

It is also called as hard superconductor & it is characterized by two critical fields H_{c1} & H_{c2} ($H_{c1} < H_c < H_{c2}$). It exists in three states → superconducting, mixed & normal.

1) Superconducting state:

- It occurs upto critical field H_{c1} .
- Magnetization increases with the applied magnetic field & the external magnetic flux is completely expelled from the interior of the material.

2) Mixed state:-

- This region extends from H_{c1} to H_{c2} .
- At H_{c1} , the magnetic flux penetrates the material.

- Between H_{c1} & H_{c2} , the material is in a mixed state magnetically but electrically it is a superconductor.

- Meissner effect is incomplete.

- In this region, the superconductor is threaded by flux lines & is said to be in a vortex state.

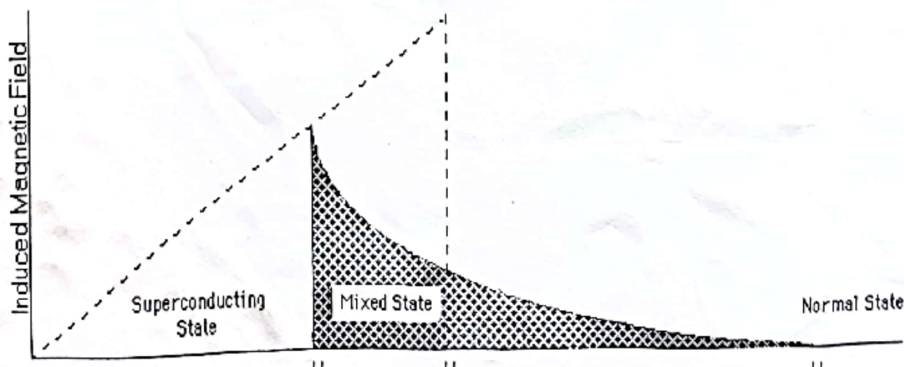
- Value of H_{c2} may be 100 times higher than H_{c1} .

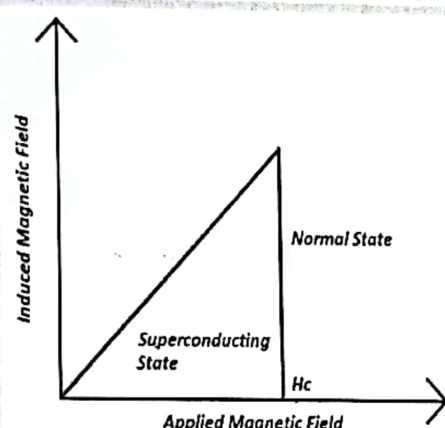
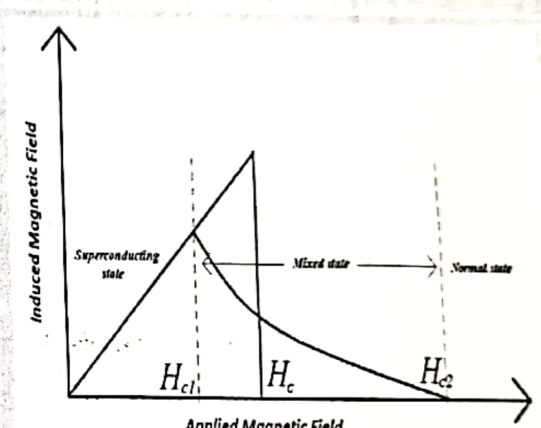
3) Normal state:

When the magnetic field exceeds critical field strength H_{c2} , magnetization vanishes completely. The sample is penetrated by the external field & superconductivity is destroyed. The specimen reverts from superconducting state to normal state.

Fig. 10

Type II Superconductor



Sr	Property	Type-I Superconductor	Type-II Superconductor
1	Meissner effect	They exhibit complete Meissner effect.	They do not exhibit Meissner effect completely.
2	Diamagnetic Behavior	They show perfect diamagnetic behavior	They do not show perfect diamagnetic behavior
3	Critical magnetic field	They have only one critical magnetic field H_c .	They have two critical magnetic fields. At lower critical magnetic field H_{c1} flux starts penetrating the superconductor. At upper critical magnetic field H_{c2} flux enters into superconductor and it loses superconductivity.
	Variation of Magnetization (M) of superconductor with applied Magnetic Field (H)	 The graph shows the variation of induced magnetic field with applied magnetic field for a Type-I superconductor. The y-axis is labeled 'Induced Magnetic Field' and the x-axis is labeled 'Applied Magnetic Field'. The curve is zero for the 'Superconducting State' and increases linearly for the 'Normal State' at the critical field H_c.	 The graph shows the variation of induced magnetic field with applied magnetic field for a Type-II superconductor. The y-axis is labeled 'Induced Magnetic Field' and the x-axis is labeled 'Applied Magnetic Field'. The curve shows three regions: 'Superconducting state' (zero), 'Mixed state' (partial penetration), and 'Normal state' (linear increase). The critical fields H_{c1}, H_c, and H_{c2} are marked on the x-axis.
	States of the material	Material exists in two states Normal state Superconducting state	Material exists in three states for Normal state Mixed state Superconducting state
	Type	They are known as soft superconductors	They are known as hard superconductors
	Examples	Aluminum, lead, mercury, etc	Nb-Sn, Nb-Ti, Nb-Zr, Va-Ga, etc

Josephson Effect

Josephson junction: Two superconductors connected by a thin layer of insulating material is called Josephson junction.

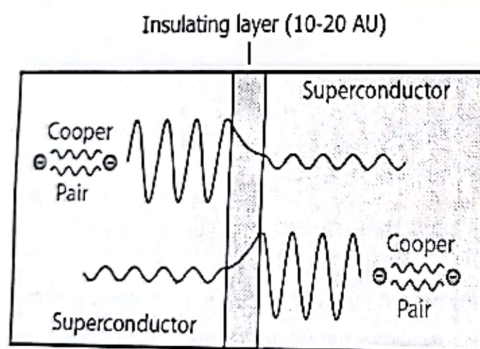
DC Josephson effect :

- When two superconductors are separated by a thin insulating layer, cooper pairs tunnel through the junction and current flows across the junction without any external applied voltage.
- If this current doesn't exceed critical current I_c , voltage across the junction is zero. This effect is known as DC Josephson effect.
- The cooper pairs can be represented by a wave function, which is the same for all pairs.
- The super-current through the junction is given by

$$I = I_c \sin \phi$$

Where, ϕ = phase difference between the wave functions describing Cooper pair

I_c = critical current at zero voltage condition.



AC Josephson effect :

- When the DC voltage is applied to the junction and current I_j through the junction exceeds a critical value I_c , a potential difference V appears across the Josephson junction and the junction current I_j varies sinusoidally with time. This effect is called ac Josephson effect.
- Energies of cooper pair on both sides of the barrier differ by $2eV$.
- Frequency ν of the associated photon will be-

$$E = h\nu = 2eV$$

$$\nu = 2(e/h)V$$

Superconducting Quantum Interference Devices (SQUID)

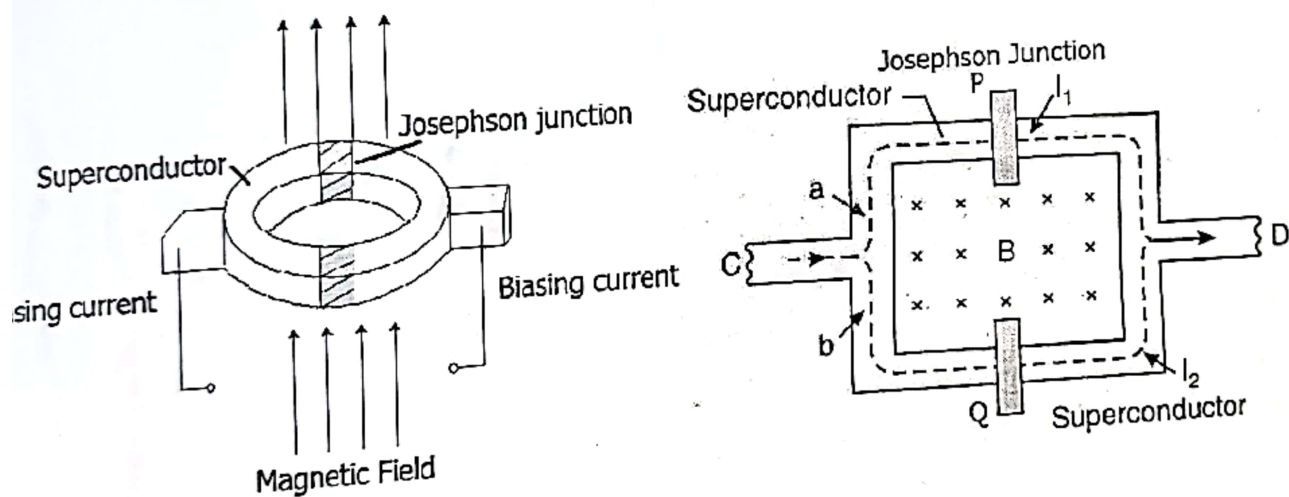
A SQUID (Superconducting Quantum Interference Device) is a very sensitive magnetometer used to measure extremely subtle (small) magnetic flux of the order of 10^{-18} Tesla. Their working is based on superconducting loops containing Josephson junctions.

Construction of SQUID

There are two main types of SQUID: direct current (DC) and radio frequency (RF). A radio frequency (RF) SQUID is made up of one Josephson junction, which is mounted on a superconducting ring. A direct current (DC) SQUID consists of two Josephson junctions in parallel, which is more sensitive.

Construction

SQUIDs are usually fabricated from lead or pure niobium. The tunnel barrier is oxidized onto lead or niobium surface. The entire device is cooled to within a few degrees of absolute zero with liquid helium. A schematic of a two-junction dc SQUID is shown in figure. It consists of two Josephson



Working

A constant dc supercurrent is applied to the SQUID. This current is known as bias current which enters into the SQUID through arm C. It is divided along the paths a and b and again merge into one and leaves through the arm D. I_1 and I_2 are currents tunneling through Josephson junctions.

2 Superconducting magnets

Conventional electromagnets are very much bigger in size. They consume large electrical power to maintain the magnetic field and also require continuous cooling. The use of superconductors in designing of magnets would have following advantages:

- **Intense magnetic field**

Superconducting magnet has ability to support a very high current density with almost no resistance. Due to this electromagnets can be constructed that generate intense magnetic fields with little electrical power input. Type-II superconductors such as niobium-titanium (NbTi) alloys, niobium-tin (Nb₃Sn) alloys can produce high magnetic fields of around 9-10 Tesla.

- **Compact and more efficient setup**

Superconducting magnet systems are quite compact and occupy a small space.

- **Can be operated for longer duration**

In the persistent mode of operation, the L/R time constant is extremely long and the magnet can be operated for days or even months at a nearly constant field.

3 Electronics

Variety of devices such as SQUIDS, transistors, ICs, etc can be designed using superconductors. Main advantage of using superconductors in designing of electronic components is reduction in heat losses and flow of current in the electronic circuit without any resistance. Some of the applications of superconductors in electronics are listed as below:

- **Josephson junctions**

Josephson junctions are used in fast electronic switches or sensitive magnetometers. A magnetometer is able to detect very small magnetic fields of the order of 0-15T.

- **Supercomputers**

The semiconductor logic elements have a speed limit. They operate at a speed in orders of few nanoseconds. Logic elements based on Josephson junction can operate at the speed of few picoseconds. Use of superconductors in logic gates will drastically increase the speed of computers.

- **SQUID (superconducting quantum interference device)**

SQUIDS are used for Non-destructive corrosion testing, Magnetoencephalography (MEG), observing neural activities inside the brain (MRI scan), study of magnetic properties of material, oil prospecting, mineral exploration, earthquake prediction, geothermal energy survey, etc.

- **Transistors**

Superconducting transistors based on Josephson junctions could be used to switch voltages very quickly. They will significantly speed up the processing of signals or data in microprocessors.

• Circuitry connections

Circuit connections can be made through superconductive films. This would have advantage that information can be transmitted more quickly without losses.

• IC fabrication

At present processing power of ICs is limited due to I^2R losses of components. Use of superconductors will make ICs more efficient.

4 Medical

Two of the properties of superconductors are extremely useful in medical field – production of extensive magnetic field in devices such as MRI (upto 1.5 Tesla) and detection of smaller magnetic fields using SQUIDs (of the order of 10^{-14} Tesla)

Magnetoencephalography (MEG)

- Magnetoencephalography (MEG) is a technique for investigating human brain activities on a millisecond basis. It can detect where in the brain activity is produced.
- Brain generates neuromagnetic signals that are extremely small (of the order of 10^{-12} Tesla). MEG scanners use SQUIDs to detect this extremely small magnetic field.

Magnetic Resonance Imagery (MRI)

- Magnetic resonance imaging (MRI), also known as nuclear magnetic resonance imaging, is a scanning technique for creating detailed images of the human body. The scan uses a strong magnetic field (of the order of 1.5 Tesla or higher) and radio waves to generate images of parts of the body that can't be seen with X-rays, CT scans or ultrasound. If conventional metal wires are used to produce such a strong magnetic field, they would heat to the extent of melting.
- Water molecules in human body contain hydrogen nuclei (protons). An MRI scanner applies a very strong magnetic field (about 0.2 to 3 teslas), which aligns the proton "spins". This alignment of spin can be detected by SQUIDs.
- MRI diagnosis is used for investigation of brain, including detection of tumors. MRI is also used for investigation of sports injuries, muscle injuries, skeletal problems, spinal injuries.

5 Transport - MagLev Trains

In conventional transportation system, friction between the wheels and ground or rail is one of the crucial elements. Due to friction there is a limitation of speed of the vehicle also it also increases wear and tear. Magnetic Levitation Trains (Maglev Trains) is a floating vehicle for land transportation. MagLev trains do not slide over the rails but float on an air cushion over a strongly magnetized track. As there is no mechanical friction, speeds upto 500 km/h can be easily achieved. The advantages of superconductors are:

- Permanent currents as high as about 700000 amperes can be passed through the superconducting coils.
- This produces a strong magnetic field of order of almost 5 Tesla enough to levitate the train.